

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

Electronics and Communication Engineering Department

Object Oriented Programming with JAVA (IIPC 213)

Mid Sem Exam-I

Duration: 50 min.

Max. Marks:15

All questions are compulsory.

Date:

Q1 You are designing a superclass called Product that includes common attributes such as productName, price, and description. The subclasses Electronics, Clothing, and Books will inherit from the Product class.

- (a) How would you design the constructors for the Product and its subclasses?
- (b) What role does the super() keyword play when constructing a Clothing object that inherits from Product?

Q2. You need to implement a method calculateDiscount() in the Product class that returns a default discount for any product. The discount logic should be specific for each product type.

- (a) How would you override this method in the Electronics, Clothing, and Books subclasses?
- (b) Describe how polymorphism allows the correct discount calculation when calling the calculateDiscount() method from a Product reference that holds a Clothing object.

Q3. You are designing a Shipping Fee Calculation system for an e-commerce platform. The shipping fee calculation differs depending on whether the customer is shipping a Domestic or International order. Additionally, the weight of the product impacts the shipping fee.

How would you overload the method calculateShippingFee() in the Shipping class to compute the fee based on the following:

- (a) For a Domestic order, calculate the fee based on the weight (double weight).
- (b) For an International order, calculate the fee based on the weight (double weight) and the destination country (String destinationCountry).